

Interprocess Communication Introduction

Interprocess Communication

Learning Objectives

- Be able to explain the purpose of Unix's key IPC mechanisms.
- Be able to identify when each IPC mechanism is appropriate.
- Be able to use IPC mechanisms to communicate across processes

Complementary Reading: [Beej's Guide to Interprocess Communication](#) Section 3, 4, and 11

Inter Process Communication

- Isolation between processes is great! It prevents bad things from happening.
- But sometimes we want to communicate!
- Today we're going to learn about *structured* communication using:
 - Signals
 - Pipes
 - Sockets
- Wednesday, we'll learn about *unstructured communication* through shared memory.

2. Signals

- Asynchronous messages that notify a process of an event.
 - "something happened"
 - NOT: "here's some details about something that happened".
- Can be sent to any process, you just have to know its PID.
- Example:
 - You want to kill the process with PID, `<pid>`. Run, `kill -9 <pid>`
- Sending signals mechanism:
 - Sending process calls `kill`.
 - (note: the naming is a misnomer, it does not always actually kill the process!)